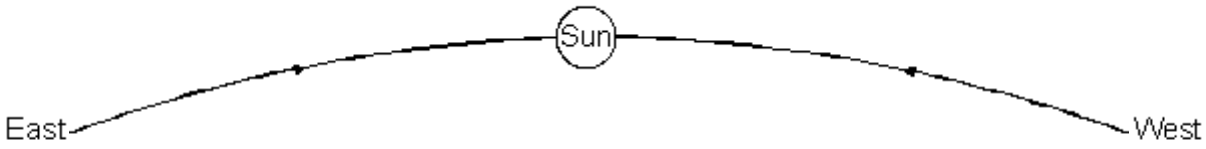
Q1. The Sun seems to move across the sky. The diagram shows part of the path of the Sun one day in winter in Britain.





(a) When the Sun is in the position shown in the diagram, what time is it?

.....

1 mark

(b) **On the diagram above** draw another line to show the path of the Sun across the sky in the summer.

1 mark

(c) The drawing below shows a sunflower plant at 8 o'clock in the morning, at 12 noon, and at 4 o'clock in the afternoon on a summer's day.

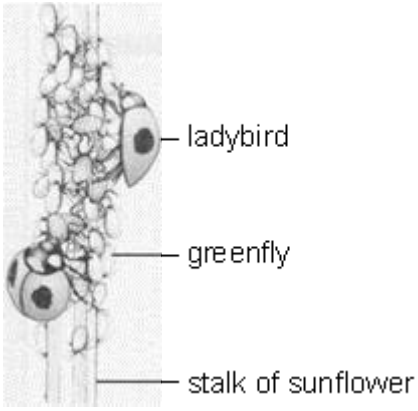


The sunflower plant bends towards the East in the morning and the West in the afternoon.
Explain why it does this.

.....

1 mark

(d) The drawing shows greenflies and ladybirds on a sunflower stalk.



Look at the food chain.

sunflower → greenfly → ladybird → bird

(i) What is the producer in this food chain?

1 mark

(ii) Which **two** living things in the food chain are predators?

..... and .....

1 mark

(iii) Why do sunflowers grow better if there are lots of ladybirds in the garden?

.....

1 mark

Maximum 6 marks

Q1.

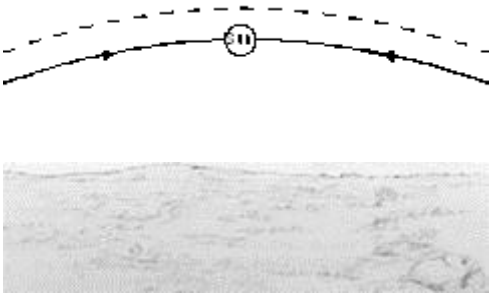
(a) 12 noon **or** midday

accept '12 o'clock'

accept answers between 11.30 and 12.30

1 (L4)

(b)



the whole of the curved path must be drawn above the printed curve but the two lines need not be parallel
the highest point must be above the position of the Sun in the diagram

*do **not** accept a horizontal or U-shaped path*

1 (L4)

(c) any **one** from

- that is where the Sun is
- it bends towards the Sun
- to get the most light

accept 'to catch the Sun's rays'
accept 'to follow the Sun'

*accept 'the Sun is in the East in the morning **or** in the West in the afternoon'*
***or** 'the Sun appears to move'*

*do **not** accept 'the Sun moves'*

*do **not** accept 'it faces the warmth of the Sun'*

1 (L3)

(d) (i) sunflower

*accept 'the plant' **or** 'flower'*

1 (L4)

(ii) ladybird and bird

answers may be in either order
***both** answers are required for the mark*

1 (L4)

(iii) any **one** from

- ladybirds eat the greenflies

accept 'they eat greenflies'

- there are fewer greenflies to eat the sunflower
  accept 'there are less greenflies'
  *do **not** accept 'greenflies damage the sunflower'*

1 (L4)

[6]